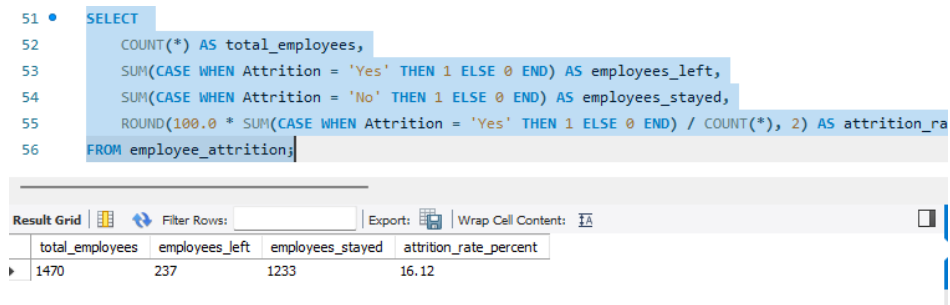


MY SQL QUERIES

PROJECT -IBM HR Analytics

1. Overall Attrition Rate

```
SELECT
    COUNT(*) AS total_employees,
    SUM(CASE WHEN Attrition = 'Yes' THEN 1 ELSE 0 END) AS employees_left,
    SUM(CASE WHEN Attrition = 'No' THEN 1 ELSE 0 END) AS employees_stayed,
    ROUND(100.0 * SUM(CASE WHEN Attrition = 'Yes' THEN 1 ELSE 0 END) / COUNT(*), 2) AS
attrition_rate_percent
FROM employee_attrition;
```



The screenshot shows a SQL query editor with the following query:

```
51 • SELECT
52     COUNT(*) AS total_employees,
53     SUM(CASE WHEN Attrition = 'Yes' THEN 1 ELSE 0 END) AS employees_left,
54     SUM(CASE WHEN Attrition = 'No' THEN 1 ELSE 0 END) AS employees_stayed,
55     ROUND(100.0 * SUM(CASE WHEN Attrition = 'Yes' THEN 1 ELSE 0 END) / COUNT(*), 2) AS attrition_ra
56 FROM employee_attrition;
```

Below the query editor is a 'Result Grid' with the following data:

total_employees	employees_left	employees_stayed	attrition_rate_percent
1470	237	1233	16.12

2. Attrition by Age Group

```
SELECT
    CASE
        WHEN Age BETWEEN 18 AND 25 THEN '18-25'
        WHEN Age BETWEEN 26 AND 35 THEN '26-35'
        WHEN Age BETWEEN 36 AND 45 THEN '36-45'
        WHEN Age BETWEEN 46 AND 55 THEN '46-55'
        ELSE '56+'
    END AS age_group,
    COUNT(*) AS total_employees,
    SUM(CASE WHEN Attrition = 'Yes' THEN 1 ELSE 0 END) AS left_count,
    ROUND(100.0 * SUM(CASE WHEN Attrition = 'Yes' THEN 1 ELSE 0 END) / COUNT(*), 2) AS
attrition_rate
FROM employee_attrition
GROUP BY age_group
ORDER BY attrition_rate DESC;
```

```

51 • SELECT
52     CASE
53         WHEN Age BETWEEN 18 AND 25 THEN '18-25'
54         WHEN Age BETWEEN 26 AND 35 THEN '26-35'
55         WHEN Age BETWEEN 36 AND 45 THEN '36-45'
56         WHEN Age BETWEEN 46 AND 55 THEN '46-55'
57         ELSE '56+'
58     END AS age_group,
59     COUNT(*) AS total_employees,

```

age_group	total_employees	left_count	attrition_rate
18-25	123	44	35.77
26-35	606	116	19.14
56+	47	8	17.02
46-55	226	26	11.50
36-45	468	43	9.19

3. Attrition by Gender

```

SELECT
    Gender,
    COUNT(*) AS total_employees,
    SUM(CASE WHEN Attrition = 'Yes' THEN 1 ELSE 0 END) AS left_count,
    ROUND(100.0 * SUM(CASE WHEN Attrition = 'Yes' THEN 1 ELSE 0 END) / COUNT(*), 2) AS
attrition_rate
FROM employee_attrition
GROUP BY Gender
ORDER BY attrition_rate DESC;

```

```

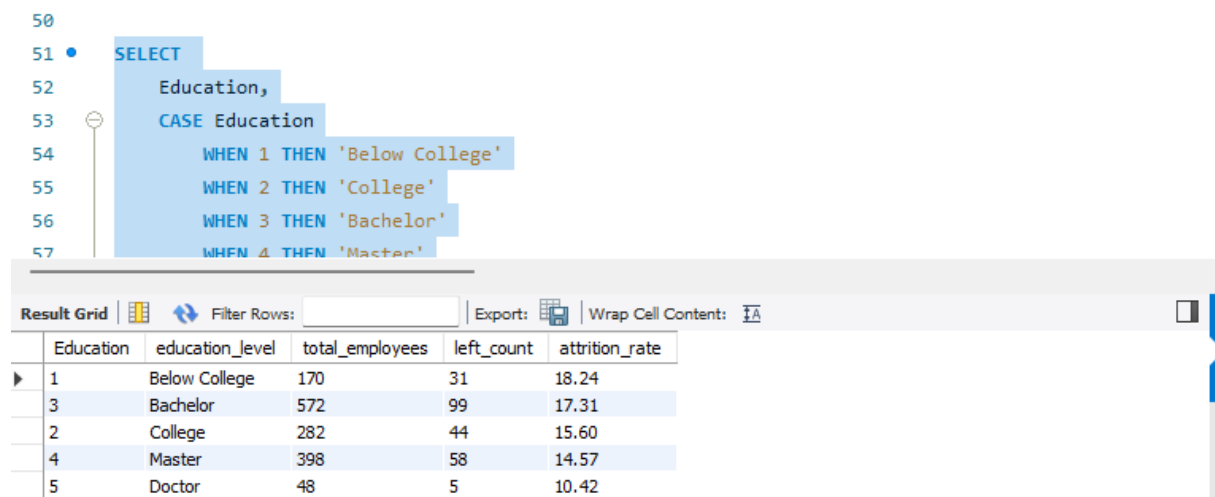
51 • SELECT
52     Gender,
53     COUNT(*) AS total_employees,
54     SUM(CASE WHEN Attrition = 'Yes' THEN 1 ELSE 0 END) AS left_count,
55     ROUND(100.0 * SUM(CASE WHEN Attrition = 'Yes' THEN 1 ELSE 0 END) / COUNT(*), 2) AS attrition_rate
56 FROM employee_attrition
57 GROUP BY Gender
58 ORDER BY attrition_rate DESC;

```

Gender	total_employees	left_count	attrition_rate
Male	882	150	17.01
Female	588	87	14.80

4. Attrition by Education Level

```
SELECT
  Education,
  CASE Education
    WHEN 1 THEN 'Below College'
    WHEN 2 THEN 'College'
    WHEN 3 THEN 'Bachelor'
    WHEN 4 THEN 'Master'
    WHEN 5 THEN 'Doctor'
  END AS education_level,
  COUNT(*) AS total_employees,
  SUM(CASE WHEN Attrition = 'Yes' THEN 1 ELSE 0 END) AS left_count,
  ROUND(100.0 * SUM(CASE WHEN Attrition = 'Yes' THEN 1 ELSE 0 END) / COUNT(*), 2) AS
attrition_rate
FROM employee_attrition
GROUP BY Education
ORDER BY attrition_rate DESC;
```



The screenshot shows a SQL IDE interface. The top part displays the SQL query for attrition by education level. Below the query, the 'Result Grid' is visible, showing the results of the query. The results are ordered by attrition_rate in descending order.

	Education	education_level	total_employees	left_count	attrition_rate
1	Below College	Below College	170	31	18.24
3	Bachelor	Bachelor	572	99	17.31
2	College	College	282	44	15.60
4	Master	Master	398	58	14.57
5	Doctor	Doctor	48	5	10.42

5. Attrition by Department

```
SELECT
  Department,
  COUNT(*) AS total_employees,
  SUM(CASE WHEN Attrition = 'Yes' THEN 1 ELSE 0 END) AS left_count,
  ROUND(100.0 * SUM(CASE WHEN Attrition = 'Yes' THEN 1 ELSE 0 END) / COUNT(*), 2) AS
attrition_rate
FROM employee_attrition
GROUP BY Department
ORDER BY attrition_rate DESC;
```

```

51 • SELECT
52     Department,
53     COUNT(*) AS total_employees,
54     SUM(CASE WHEN Attrition = 'Yes' THEN 1 ELSE 0 END) AS left_count,
55     ROUND(100.0 * SUM(CASE WHEN Attrition = 'Yes' THEN 1 ELSE 0 END) / COUNT(*), 2) AS attrition_rate
56 FROM employee_attrition
57 GROUP BY Department
58 ORDER BY attrition_rate DESC;

```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
Department	total_employees	left_count	attrition_rate
Sales	446	92	20.63
Human Resources	63	12	19.05
Research & Development	961	133	13.84

6. Attrition by Job Role

```

SELECT
    JobRole,
    COUNT(*) AS total_employees,
    SUM(CASE WHEN Attrition = 'Yes' THEN 1 ELSE 0 END) AS left_count,
    ROUND(100.0 * SUM(CASE WHEN Attrition = 'Yes' THEN 1 ELSE 0 END) / COUNT(*), 2) AS
attrition_rate
FROM employee_attrition
GROUP BY JobRole
ORDER BY attrition_rate DESC;

```

```

51 • SELECT
52     JobRole,
53     COUNT(*) AS total_employees,
54     SUM(CASE WHEN Attrition = 'Yes' THEN 1 ELSE 0 END) AS left_count,
55     ROUND(100.0 * SUM(CASE WHEN Attrition = 'Yes' THEN 1 ELSE 0 END) / COUNT(*), 2) AS attrition_rate
56 FROM employee_attrition
57 GROUP BY JobRole
58 ORDER BY attrition_rate DESC;

```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
JobRole	total_employees	left_count	attrition_rate
Sales Representative	83	33	39.76
Laboratory Technician	259	62	23.94
Human Resources	52	12	23.08
Sales Executive	326	57	17.48
Research Scientist	292	47	16.10

7. Attrition by Marital Status

```
SELECT
    MaritalStatus,
    COUNT(*) AS total_employees,
    SUM(CASE WHEN Attrition = 'Yes' THEN 1 ELSE 0 END) AS left_count,
    ROUND(100.0 * SUM(CASE WHEN Attrition = 'Yes' THEN 1 ELSE 0 END) / COUNT(*), 2) AS
attrition_rate
FROM employee_attrition
GROUP BY MaritalStatus
ORDER BY attrition_rate DESC;
```

```
51 • SELECT
52     MaritalStatus,
53     COUNT(*) AS total_employees,
54     SUM(CASE WHEN Attrition = 'Yes' THEN 1 ELSE 0 END) AS left_count,
55     ROUND(100.0 * SUM(CASE WHEN Attrition = 'Yes' THEN 1 ELSE 0 END) / COUNT(*), 2) AS attrition_rate
56 FROM employee_attrition
57 GROUP BY MaritalStatus
58 ORDER BY attrition_rate DESC;
```

Result Grid Filter Rows: Export: Wrap Cell Content:				
MaritalStatus	total_employees	left_count	attrition_rate	
Single	470	120	25.53	
Married	673	84	12.48	
Divorced	327	33	10.09	

8. Monthly Income Comparison (Attrition vs Non-Attrition)

```
SELECT
    Attrition,
    COUNT(*) AS employee_count,
    ROUND(AVG(MonthlyIncome), 2) AS avg_monthly_income,
    ROUND(MIN(MonthlyIncome), 2) AS min_income,
    ROUND(MAX(MonthlyIncome), 2) AS max_income
FROM employee_attrition
GROUP BY Attrition;
```

```

51 • SELECT
52     Attrition,
53     COUNT(*) AS employee_count,
54     ROUND(AVG(MonthlyIncome), 2) AS avg_monthly_income,
55     ROUND(MIN(MonthlyIncome), 2) AS min_income,
56     ROUND(MAX(MonthlyIncome), 2) AS max_income
57 FROM employee_attrition
58 GROUP BY Attrition;

```

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

	Attrition	employee_count	avg_monthly_income	min_income	max_income
Yes	237	4787.09	1009	19859	
No	1233	6832.74	1051	19999	

9. Average Monthly Income by Education and Attrition

```

SELECT
CASE Education
    WHEN 1 THEN 'Below College'
    WHEN 2 THEN 'College'
    WHEN 3 THEN 'Bachelor'
    WHEN 4 THEN 'Master'
    WHEN 5 THEN 'Doctor'
END AS education_level,
Attrition,
ROUND(AVG(MonthlyIncome), 2) AS avg_monthly_income,
COUNT(*) AS employee_count
FROM employee_attrition
GROUP BY Education, Attrition
ORDER BY Education, Attrition;

```

```

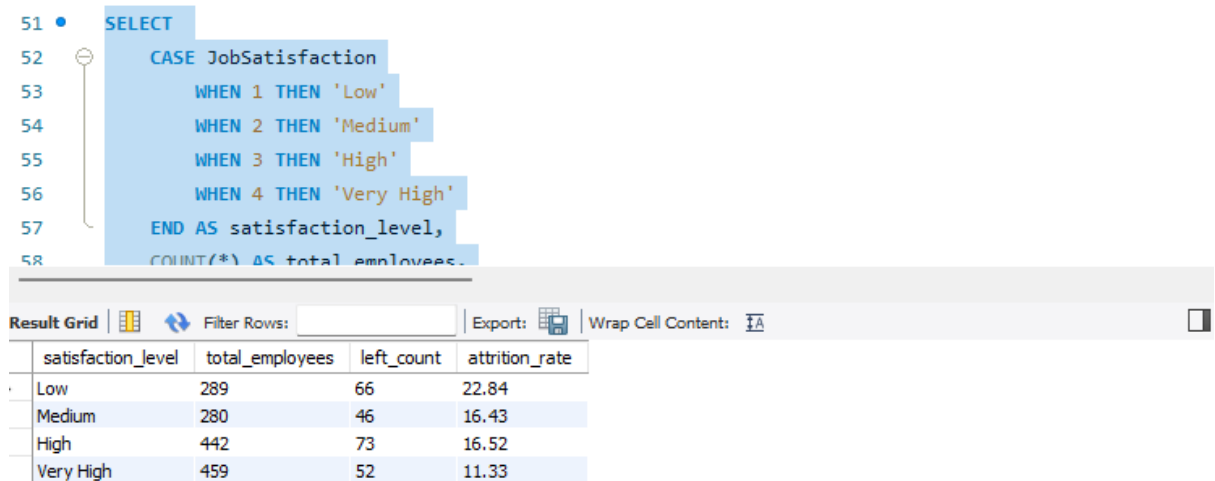
51 • SELECT
52     CASE Education
53         WHEN 1 THEN 'Below College'
54         WHEN 2 THEN 'College'
55         WHEN 3 THEN 'Bachelor'
56         WHEN 4 THEN 'Master'
57         WHEN 5 THEN 'Doctor'
58     END AS education_level,
59     Attrition,

```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
education_level	Attrition	avg_monthly_income	employee_count
Below College	No	5926.13	139
Below College	Yes	4360.16	31
College	No	6586.06	238
College	Yes	4282.55	44
Bachelor	No	6882.92	473

10. Job Satisfaction Impact on Attrition

```
SELECT
  CASE JobSatisfaction
    WHEN 1 THEN 'Low'
    WHEN 2 THEN 'Medium'
    WHEN 3 THEN 'High'
    WHEN 4 THEN 'Very High'
  END AS satisfaction_level,
  COUNT(*) AS total_employees,
  SUM(CASE WHEN Attrition = 'Yes' THEN 1 ELSE 0 END) AS left_count,
  ROUND(100.0 * SUM(CASE WHEN Attrition = 'Yes' THEN 1 ELSE 0 END) / COUNT(*), 2) AS
attrition_rate
FROM employee_attrition
GROUP BY JobSatisfaction
ORDER BY JobSatisfaction;
```



The screenshot shows a SQL query editor with the following code:

```
51 SELECT
52 CASE JobSatisfaction
53 WHEN 1 THEN 'Low'
54 WHEN 2 THEN 'Medium'
55 WHEN 3 THEN 'High'
56 WHEN 4 THEN 'Very High'
57 END AS satisfaction_level,
58 COUNT(*) AS total_employees,
```

Below the editor is a 'Result Grid' with the following data:

satisfaction_level	total_employees	left_count	attrition_rate
Low	289	66	22.84
Medium	280	46	16.43
High	442	73	16.52
Very High	459	52	11.33

11. Work-Life Balance Impact

```
SELECT
  CASE WorkLifeBalance
    WHEN 1 THEN 'Bad'
    WHEN 2 THEN 'Good'
    WHEN 3 THEN 'Better'
    WHEN 4 THEN 'Best'
  END AS wlb_level,
  COUNT(*) AS total_employees,
  SUM(CASE WHEN Attrition = 'Yes' THEN 1 ELSE 0 END) AS left_count,
  ROUND(100.0 * SUM(CASE WHEN Attrition = 'Yes' THEN 1 ELSE 0 END) / COUNT(*), 2) AS
attrition_rate
FROM employee_attrition
GROUP BY WorkLifeBalance
ORDER BY WorkLifeBalance;
```

```

51 • SELECT
52     CASE WorkLifeBalance
53     WHEN 1 THEN 'Bad'
54     WHEN 2 THEN 'Good'
55     WHEN 3 THEN 'Better'
56     WHEN 4 THEN 'Best'
57     END AS wlb_level,
58     COUNT(*) AS total_employees.

```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
wlb_level	total_employees	left_count	attrition_rate
Bad	80	25	31.25
Good	344	58	16.86
Better	893	127	14.22
Best	153	27	17.65

12. Overtime Impact

```

SELECT
    OverTime,
    COUNT(*) AS total_employees,
    SUM(CASE WHEN Attrition = 'Yes' THEN 1 ELSE 0 END) AS left_count,
    ROUND(100.0 * SUM(CASE WHEN Attrition = 'Yes' THEN 1 ELSE 0 END) / COUNT(*), 2) AS
attrition_rate
FROM employee_attrition
GROUP BY OverTime;

```

```

51 • SELECT
52     OverTime,
53     COUNT(*) AS total_employees,
54     SUM(CASE WHEN Attrition = 'Yes' THEN 1 ELSE 0 END) AS left_count,
55     ROUND(100.0 * SUM(CASE WHEN Attrition = 'Yes' THEN 1 ELSE 0 END) / COUNT(*), 2) AS attrition_rate
56 FROM employee_attrition
57 GROUP BY OverTime;

```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
OverTime	total_employees	left_count	attrition_rate
Yes	416	127	30.53
No	1054	110	10.44

13. Environment Satisfaction Impact

```

SELECT
    CASE EnvironmentSatisfaction
    WHEN 1 THEN 'Low'
    WHEN 2 THEN 'Medium'
    WHEN 3 THEN 'High'
    WHEN 4 THEN 'Very High'
    END AS env_satisfaction,
    COUNT(*) AS total_employees,
    SUM(CASE WHEN Attrition = 'Yes' THEN 1 ELSE 0 END) AS left_count,

```



```

ROUND(100.0 * SUM(CASE WHEN Attrition = 'Yes' THEN 1 ELSE 0 END) / COUNT(*), 2) AS
attrition_rate
FROM employee_attrition
GROUP BY EnvironmentSatisfaction
ORDER BY EnvironmentSatisfaction;

```

51 • **SELECT**

52 CASE EnvironmentSatisfaction

53 WHEN 1 THEN 'Low'

54 WHEN 2 THEN 'Medium'

55 WHEN 3 THEN 'High'

56 WHEN 4 THEN 'Very High'

57 END AS env_satisfaction,

58 COUNT(*) AS total_employees,

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

env_satisfaction	total_employees	left_count	attrition_rate
Low	284	72	25.35
Medium	287	43	14.98
High	453	62	13.69
Very High	446	60	13.45

14. Job Involvement Impact

```

SELECT
CASE JobInvolvement
WHEN 1 THEN 'Low'
WHEN 2 THEN 'Medium'
WHEN 3 THEN 'High'
WHEN 4 THEN 'Very High'
END AS involvement_level,
COUNT(*) AS total_employees,
SUM(CASE WHEN Attrition = 'Yes' THEN 1 ELSE 0 END) AS left_count,
ROUND(100.0 * SUM(CASE WHEN Attrition = 'Yes' THEN 1 ELSE 0 END) / COUNT(*), 2) AS
attrition_rate
FROM employee_attrition
GROUP BY JobInvolvement
ORDER BY JobInvolvement;

```

51 • **SELECT**

52 CASE JobInvolvement

53 WHEN 1 THEN 'Low'

54 WHEN 2 THEN 'Medium'

55 WHEN 3 THEN 'High'

56 WHEN 4 THEN 'Very High'

57 END AS involvement_level,

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

involvement_level	total_employees	left_count	attrition_rate
Low	83	28	33.73
Medium	375	71	18.93
High	868	125	14.40
Very High	144	13	9.03

15. Percent Salary Hike Impact

```
SELECT
    Attrition,
    ROUND(AVG(PercentSalaryHike), 2) AS avg_salary_hike,
    ROUND(MIN(PercentSalaryHike), 2) AS min_hike,
    ROUND(MAX(PercentSalaryHike), 2) AS max_hike
FROM employee_attrition
GROUP BY Attrition;
```

```
51 • SELECT
52     Attrition,
53     ROUND(AVG(PercentSalaryHike), 2) AS avg_salary_hike,
54     ROUND(MIN(PercentSalaryHike), 2) AS min_hike,
55     ROUND(MAX(PercentSalaryHike), 2) AS max_hike
56 FROM employee_attrition
57 GROUP BY Attrition;
```

Result Grid				
		Filter Rows:	Export:	Wrap Cell Content:
Attrition	avg_salary_hike	min_hike	max_hike	
Yes	15.10	11	25	
No	15.23	11	25	

16. Stock Option Level Impact

```
SELECT
    StockOptionLevel,
    COUNT(*) AS total_employees,
    SUM(CASE WHEN Attrition = 'Yes' THEN 1 ELSE 0 END) AS left_count,
    ROUND(100.0 * SUM(CASE WHEN Attrition = 'Yes' THEN 1 ELSE 0 END) / COUNT(*), 2) AS
attrition_rate
FROM employee_attrition
GROUP BY StockOptionLevel
ORDER BY StockOptionLevel;
```

```
51 • SELECT
52     StockOptionLevel,
53     COUNT(*) AS total_employees,
54     SUM(CASE WHEN Attrition = 'Yes' THEN 1 ELSE 0 END) AS left_count,
55     ROUND(100.0 * SUM(CASE WHEN Attrition = 'Yes' THEN 1 ELSE 0 END) / COUNT(*), 2) AS attrition_rate
56 FROM employee_attrition
57 GROUP BY StockOptionLevel
58 ORDER BY StockOptionLevel;
```

Result Grid				
		Filter Rows:	Export:	Wrap Cell Content:
StockOptionLevel	total_employees	left_count	attrition_rate	
0	631	154	24.41	
1	596	56	9.40	
2	158	12	7.59	
3	85	15	17.65	

17. Breakdown of Distance From Home by Job Role and Attrition

```
SELECT
    JobRole,
    Attrition,
    ROUND(AVG(DistanceFromHome), 2) AS avg_distance,
    ROUND(MIN(DistanceFromHome), 2) AS min_distance,
    ROUND(MAX(DistanceFromHome), 2) AS max_distance,
    COUNT(*) AS employee_count
FROM employee_attrition
GROUP BY JobRole, Attrition
ORDER BY JobRole, Attrition;
```

```
i1 • SELECT
i2     JobRole,
i3     Attrition,
i4     ROUND(AVG(DistanceFromHome), 2) AS avg_distance,
i5     ROUND(MIN(DistanceFromHome), 2) AS min_distance,
i6     ROUND(MAX(DistanceFromHome), 2) AS max_distance,
i7     COUNT(*) AS employee_count
i8     FROM employee_attrition
```

JobRole	Attrition	avg_distance	min_distance	max_distance	employee_count
Healthcare Representative	No	9.20	1	29	122
Healthcare Representative	Yes	17.67	2	29	9
Human Resources	No	6.60	1	26	40
Human Resources	Yes	13.42	1	23	12
Laboratory Technician	No	9.33	1	29	197

18. Compare Average Monthly Income by Education and Attrition

```
SELECT
    CASE Education
        WHEN 1 THEN 'Below College'
        WHEN 2 THEN 'College'
        WHEN 3 THEN 'Bachelor'
        WHEN 4 THEN 'Master'
        WHEN 5 THEN 'Doctor'
    END AS education_level,
    Attrition,
    ROUND(AVG(MonthlyIncome), 2) AS avg_monthly_income,
    ROUND(MIN(MonthlyIncome), 2) AS min_income,
    ROUND(MAX(MonthlyIncome), 2) AS max_income
FROM employee_attrition
GROUP BY Education, Attrition
ORDER BY Education, Attrition;
```

```

50
51 • SELECT
52     CASE Education
53         WHEN 1 THEN 'Below College'
54         WHEN 2 THEN 'College'
55         WHEN 3 THEN 'Bachelor'
56         WHEN 4 THEN 'Master'
57         WHEN 5 THEN 'Doctor'

```

education_level	Attrition	avg_monthly_income	min_income	max_income
Below College	No	5926.13	1052	19973
Below College	Yes	4360.16	1009	19845
College	No	6586.06	1051	19613
College	Yes	4282.55	1569	10482
Bachelor	No	6882.92	1200	19926

19. Years at Company Analysis

```

SELECT
    Attrition,
    ROUND(AVG(YearsAtCompany), 2) AS avg_years_at_company,
    ROUND(MIN(YearsAtCompany), 2) AS min_years,
    ROUND(MAX(YearsAtCompany), 2) AS max_years
FROM employee_attrition
GROUP BY Attrition;

```

```

51 • SELECT
52     Attrition,
53     ROUND(AVG(YearsAtCompany), 2) AS avg_years_at_company,
54     ROUND(MIN(YearsAtCompany), 2) AS min_years,
55     ROUND(MAX(YearsAtCompany), 2) AS max_years
56 FROM employee_attrition
57 GROUP BY Attrition;

```

Attrition	avg_years_at_company	min_years	max_years
Yes	5.13	0	40
No	7.37	0	37

20. Years Since Last Promotion Impact

```

SELECT
    promotion_gap,
    Attrition,
    employee_count,
    ROUND(100.0 * employee_count / SUM(employee_count) OVER (PARTITION BY
promotion_gap), 2) AS percentage
FROM (
    SELECT
        CASE

```

```

        WHEN YearsSinceLastPromotion = 0 THEN 'No promotion'
        WHEN YearsSinceLastPromotion BETWEEN 1 AND 2 THEN '1-2 years'
        WHEN YearsSinceLastPromotion BETWEEN 3 AND 5 THEN '3-5 years'
        ELSE '5+ years'
    END AS promotion_gap,
    Attrition,
    COUNT(*) AS employee_count
FROM employee_attrition
GROUP BY promotion_gap, Attrition
) AS subquery
ORDER BY promotion_gap, Attrition;

```

```

51 • SELECT
52     promotion_gap,
53     Attrition,
54     employee_count,
55     ROUND(100.0 * employee_count / SUM(employee_count) OVER (PARTITION BY promotion_gap), 2) AS percentage
56 FROM (
57     SELECT
58     CASE

```

promotion_gap	Attrition	employee_count	percentage
1-2 years	No	440	85.27
1-2 years	Yes	76	14.73
3-5 years	No	142	89.87
3-5 years	Yes	16	10.13
5+ years	No	180	83.72

21. Number of Companies Worked Impact

```

SELECT
    NumCompaniesWorked,
    COUNT(*) AS total_employees,
    SUM(CASE WHEN Attrition = 'Yes' THEN 1 ELSE 0 END) AS left_count,
    ROUND(100.0 * SUM(CASE WHEN Attrition = 'Yes' THEN 1 ELSE 0 END) / COUNT(*), 2) AS
attrition_rate
FROM employee_attrition
GROUP BY NumCompaniesWorked
ORDER BY NumCompaniesWorked;

```

```

51 • SELECT
52     NumCompaniesWorked,
53     COUNT(*) AS total_employees,
54     SUM(CASE WHEN Attrition = 'Yes' THEN 1 ELSE 0 END) AS left_count,
55     ROUND(100.0 * SUM(CASE WHEN Attrition = 'Yes' THEN 1 ELSE 0 END) / COUNT(*), 2) AS attrition_rate
56 FROM employee_attrition
57 GROUP BY NumCompaniesWorked

```

NumCompaniesWorked	total_employees	left_count	attrition_rate
0	197	23	11.68
1	521	98	18.81
2	146	16	10.96
3	159	16	10.06
4	139	17	12.23

22. Distance From Home Impact

```
SELECT
  CASE
    WHEN DistanceFromHome <= 5 THEN '0-5 miles'
    WHEN DistanceFromHome BETWEEN 6 AND 15 THEN '6-15 miles'
    WHEN DistanceFromHome BETWEEN 16 AND 25 THEN '16-25 miles'
    ELSE '25+ miles'
  END AS distance_category,
  COUNT(*) AS total_employees,
  SUM(CASE WHEN Attrition = 'Yes' THEN 1 ELSE 0 END) AS left_count,
  ROUND(100.0 * SUM(CASE WHEN Attrition = 'Yes' THEN 1 ELSE 0 END) / COUNT(*), 2) AS
attrition_rate
FROM employee_attrition
GROUP BY distance_category
ORDER BY distance_category;
```

```
51 • SELECT
52   CASE
53     WHEN DistanceFromHome <= 5 THEN '0-5 miles'
54     WHEN DistanceFromHome BETWEEN 6 AND 15 THEN '6-15 miles'
55     WHEN DistanceFromHome BETWEEN 16 AND 25 THEN '16-25 miles'
56     ELSE '25+ miles'
57   END AS distance_category,
58   COUNT(*) AS total_employees,
```

distance_category	total_employees	left_count	attrition_rate
0-5 miles	632	87	13.77
16-25 miles	242	55	22.73
25+ miles	87	13	14.94
6-15 miles	509	82	16.11

23. Monthly Income Distribution by Department and Attrition

```
SELECT
  Department,
  Attrition,
  ROUND(AVG(MonthlyIncome), 2) AS avg_income,
  ROUND(MIN(MonthlyIncome), 2) AS min_income,
  ROUND(MAX(MonthlyIncome), 2) AS max_income,
  COUNT(*) AS employee_count
FROM employee_attrition
GROUP BY Department, Attrition
ORDER BY Department, Attrition;
```

```

51 • SELECT
52     Department,
53     Attrition,
54     ROUND(AVG(MonthlyIncome), 2) AS avg_income,
55     ROUND(MIN(MonthlyIncome), 2) AS min_income,
56     ROUND(MAX(MonthlyIncome), 2) AS max_income,
57     COUNT(*) AS employee_count

```

Department	Attrition	avg_income	min_income	max_income	employee_count
Human Resources	No	7345.98	2064	19717	51
Human Resources	Yes	3715.75	1555	10482	12
Research & Development	No	6630.33	1051	19999	828
Research & Development	Yes	4108.08	1009	19859	133
Sales	No	7232.24	1052	19847	354

24. Performance Rating vs Attrition

```

SELECT
    CASE PerformanceRating
        WHEN 1 THEN 'Low'
        WHEN 2 THEN 'Good'
        WHEN 3 THEN 'Excellent'
        WHEN 4 THEN 'Outstanding'
    END AS rating,
    COUNT(*) AS total_employees,
    SUM(CASE WHEN Attrition = 'Yes' THEN 1 ELSE 0 END) AS left_count,
    ROUND(100.0 * SUM(CASE WHEN Attrition = 'Yes' THEN 1 ELSE 0 END) / COUNT(*), 2) AS
attrition_rate
FROM employee_attrition
GROUP BY PerformanceRating
ORDER BY PerformanceRating;

```

```

51 • SELECT
52     CASE PerformanceRating
53         WHEN 1 THEN 'Low'
54         WHEN 2 THEN 'Good'
55         WHEN 3 THEN 'Excellent'
56         WHEN 4 THEN 'Outstanding'
57     END AS rating,
58     COUNT(*) AS total_employees,
59     SUM(CASE WHEN Attrition = 'Yes' THEN 1 ELSE 0 END) AS left_count,

```

rating	total_employees	left_count	attrition_rate
Excellent	1244	200	16.08
Outstanding	226	37	16.37

25. Complete Summary Statistics by Attrition

```
SELECT
  'Total Employees' AS metric,
  CAST(COUNT(*) AS CHAR) AS stayed,
  CAST(SUM(CASE WHEN Attrition = 'Yes' THEN 1 ELSE 0 END) AS CHAR) AS left_value,
  CAST(ROUND(100.0 * SUM(CASE WHEN Attrition = 'Yes' THEN 1 ELSE 0 END) / COUNT(*), 2)
AS CHAR) AS left_percent
FROM employee_attrition
```

UNION ALL

```
SELECT
  'Avg Age' AS metric,
  CAST(ROUND(AVG(CASE WHEN Attrition = 'No' THEN Age END), 1) AS CHAR),
  CAST(ROUND(AVG(CASE WHEN Attrition = 'Yes' THEN Age END), 1) AS CHAR),
  '_'
FROM employee_attrition
```

UNION ALL

```
SELECT
  'Avg Monthly Income' AS metric,
  CAST(ROUND(AVG(CASE WHEN Attrition = 'No' THEN MonthlyIncome END), 0) AS CHAR),
  CAST(ROUND(AVG(CASE WHEN Attrition = 'Yes' THEN MonthlyIncome END), 0) AS CHAR),
  '_'
FROM employee_attrition
```

UNION ALL

```
SELECT
  'Avg Years at Company' AS metric,
  CAST(ROUND(AVG(CASE WHEN Attrition = 'No' THEN YearsAtCompany END), 1) AS CHAR),
  CAST(ROUND(AVG(CASE WHEN Attrition = 'Yes' THEN YearsAtCompany END), 1) AS CHAR),
  '_'
FROM employee_attrition
```

UNION ALL

```
SELECT
  'Avg Job Satisfaction' AS metric,
  CAST(ROUND(AVG(CASE WHEN Attrition = 'No' THEN JobSatisfaction END), 2) AS CHAR),
  CAST(ROUND(AVG(CASE WHEN Attrition = 'Yes' THEN JobSatisfaction END), 2) AS CHAR),
  '_'
FROM employee_attrition
```

UNION ALL


```

SELECT
    'Avg Work Life Balance' AS metric,
    CAST(ROUND(AVG(CASE WHEN Attrition = 'No' THEN WorkLifeBalance END), 2) AS CHAR),
    CAST(ROUND(AVG(CASE WHEN Attrition = 'Yes' THEN WorkLifeBalance END), 2) AS CHAR),
    '_'
FROM employee_attrition

```

UNION ALL

```

SELECT
    '% Overtime' AS metric,
    CAST(ROUND(100.0 * SUM(CASE WHEN Attrition = 'No' AND OverTime = 'Yes' THEN 1 ELSE 0
END) / NULLIF(SUM(CASE WHEN Attrition = 'No' THEN 1 ELSE 0 END), 0), 2) AS CHAR),
    CAST(ROUND(100.0 * SUM(CASE WHEN Attrition = 'Yes' AND OverTime = 'Yes' THEN 1 ELSE 0
END) / NULLIF(SUM(CASE WHEN Attrition = 'Yes' THEN 1 ELSE 0 END), 0), 2) AS CHAR),
    '_'
FROM employee_attrition;

```

```

51 • SELECT
52     'Total Employees' AS metric,
53     CAST(COUNT(*) AS CHAR) AS stayed,
54     CAST(SUM(CASE WHEN Attrition = 'Yes' THEN 1 ELSE 0 END) AS CHAR) AS left_value,
55     CAST(ROUND(100.0 * SUM(CASE WHEN Attrition = 'Yes' THEN 1 ELSE 0 END) / COUNT(*), 2) AS CHAR) AS left_percent
56 FROM employee_attrition
57
58 UNION ALL

```

Result Grid Filter Rows: Export: Wrap Cell Content:			
metric	stayed	left_value	left_percent
Total Employees	1470	237	16.12
Avg Age	37.6	33.6	-
Avg Monthly Income	6833	4787	-
Avg Years at Company	7.4	5.1	-
Avg Job Satisfaction	2.78	2.47	-

26. High Risk Employees (Based on multiple factors)

```

SELECT
    EmployeeNumber,
    Age,
    Department,
    JobRole,
    MaritalStatus,
    OverTime,
    YearsAtCompany,
    JobSatisfaction,
    WorkLifeBalance,
    MonthlyIncome,
    CASE

```

```

        WHEN OverTime = 'Yes' AND JobSatisfaction <= 2 THEN 'CRITICAL'
        WHEN OverTime = 'Yes' OR JobSatisfaction <= 2 THEN 'HIGH'
        WHEN YearsAtCompany <= 2 THEN 'MEDIUM'
        ELSE 'LOW'
    END AS risk_level
FROM employee_attrition
WHERE Attrition = 'No' -- Currently employed but at risk
    AND (OverTime = 'Yes' OR JobSatisfaction <= 2 OR YearsAtCompany <= 2)
ORDER BY
    CASE
        WHEN OverTime = 'Yes' AND JobSatisfaction <= 2 THEN 1
        WHEN OverTime = 'Yes' OR JobSatisfaction <= 2 THEN 2
        ELSE 3
    END;

```

• **SELECT**

EmployeeNumber,

Age,

Department,

JobRole,

MaritalStatus,

OverTime.

EmployeeNumber	Age	Department	JobRole	MaritalStatus	OverTime	YearsAtCompany	JobSatisfac
10	59	Research & Development	Laboratory Technician	Married	Yes	1	1
21	32	Research & Development	Research Scientist	Divorced	Yes	6	2
51	36	Research & Development	Research Scientist	Married	Yes	1	1
58	44	Sales	Sales Executive	Divorced	Yes	4	1

27. Department-wise Attrition Risk Factors

```

SELECT
    Department,
    ROUND(AVG(CASE WHEN Attrition = 'Yes' THEN JobSatisfaction END), 2) AS
avg_satisfaction_leavers,
    ROUND(AVG(CASE WHEN Attrition = 'No' THEN JobSatisfaction END), 2) AS
avg_satisfaction_stayers,
    ROUND(100.0 * SUM(CASE WHEN OverTime = 'Yes' AND Attrition = 'Yes' THEN 1 ELSE 0 END) /
    NULLIF(SUM(CASE WHEN Attrition = 'Yes' THEN 1 ELSE 0 END), 0), 2) AS
pct_overtime_leavers,
    ROUND(100.0 * SUM(CASE WHEN OverTime = 'Yes' AND Attrition = 'No' THEN 1 ELSE 0 END) /
    NULLIF(SUM(CASE WHEN Attrition = 'No' THEN 1 ELSE 0 END), 0), 2) AS
pct_overtime_stayers
FROM employee_attrition
GROUP BY Department;

```

```

51 • SELECT
52     Department,
53     ROUND(AVG(CASE WHEN Attrition = 'Yes' THEN JobSatisfaction END), 2) AS avg_satisfaction_leavers,
54     ROUND(AVG(CASE WHEN Attrition = 'No' THEN JobSatisfaction END), 2) AS avg_satisfaction_stayers,
55     ROUND(100.0 * SUM(CASE WHEN OverTime = 'Yes' AND Attrition = 'Yes' THEN 1 ELSE 0 END) /
56         NULLIF(SUM(CASE WHEN Attrition = 'Yes' THEN 1 ELSE 0 END), 0), 2) AS pct_overtime_leavers,
57     ROUND(100.0 * SUM(CASE WHEN OverTime = 'Yes' AND Attrition = 'No' THEN 1 ELSE 0 END) /
58         NULLIF(SUM(CASE WHEN Attrition = 'No' THEN 1 ELSE 0 END), 0), 2) AS pct_overtime_stayers

```

Result Grid  Filter Rows: Export:  Wrap Cell Content: 				
Department	avg_satisfaction_leavers	avg_satisfaction_stayers	pct_overtime_leavers	pct_overtime_stayers
Sales	2.52	2.81	52.17	22.60
Research & Development	2.46	2.77	55.64	23.79
Human Resources	2.17	2.71	41.67	23.53